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Socket Group 52
Socket Project Milestone

1.
This milestone implements the initial construction of a distributed hash table (DHT). A manager registers peers and selects members for a DHT. The selected peers form a logical ring, and the leader distributes storm records from details-1950.csv among them.

Each peer stores records assigned to it in a local hash table. Records travel through the ring to reach their destination. After the leader confirms that all records were stored, it reports setup completion to the manager.

This milestone implements peer registration, DHT setup, record distribution, and setup completion. Query processing, peer joining and leaving, DHT teardown, and deregistration are outside this milestone's implementation.

The application has two programs:

- Manager: Runs continuously, listens on its configured UDP port, stores registered peer information, and processes manager commands.
- Peer: Reads commands from standard input and communicates with the manager and other peers using UDP.

The manager stores registered peers in an unordered_map keyed by peer name. Each entry contains the peer's IP address, manager and peer ports, and state: Free, Leader, or InDHT.

Each peer maintains a DhtContext containing its peer ID, ring size, peer addresses, right neighbor, hash-table size, local records, and setup progress counters. A storm record stores its 14 CSV fields and its event ID. The local hash table is represented by an unordered_map keyed by the record's computed position.

This here describes the overall history of the commit timelines, including overall descriptions in what work was done with each commit:

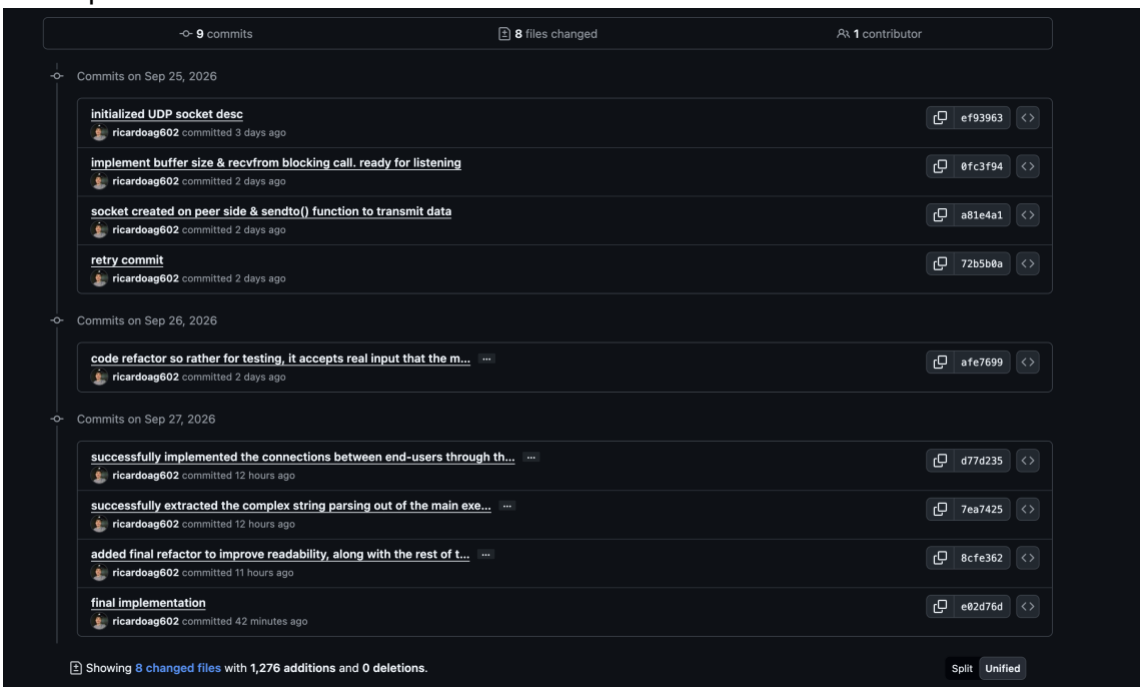
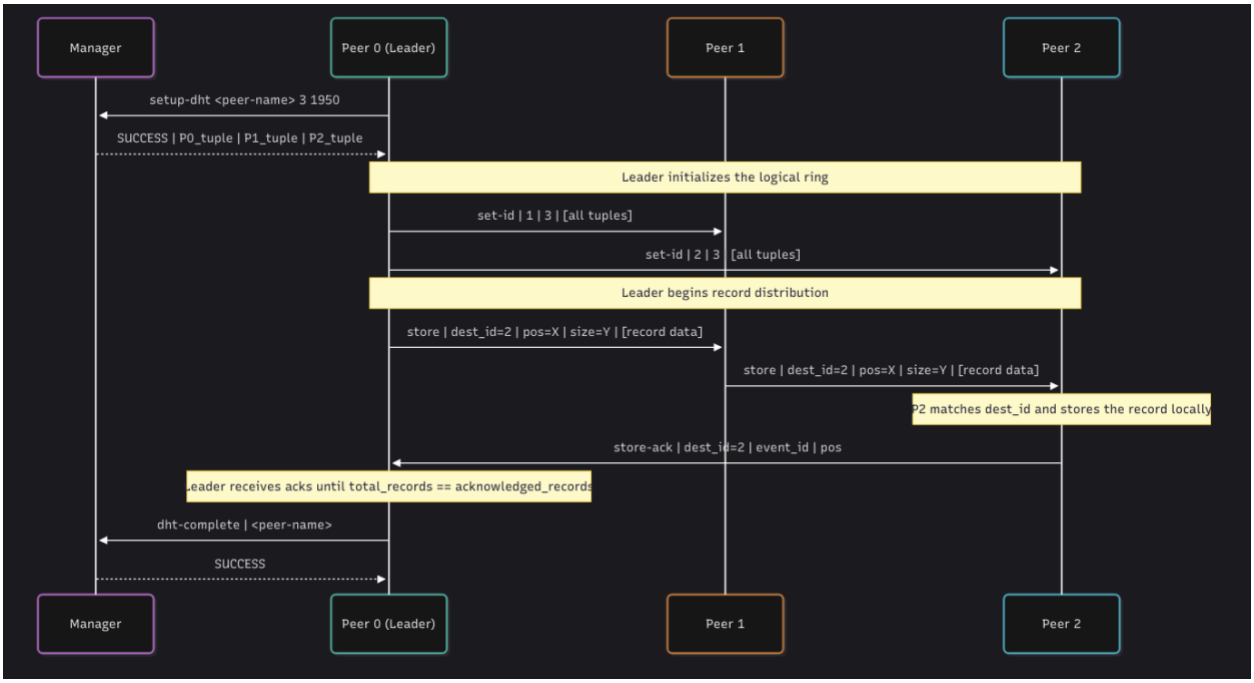


diagram of design:



Here is a link to the repository & video:
https://github.com/ricardoag602/dht-project.git